

Activity # 16: Basic Packet Analysis and Filtering

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PART 1 — IP Address Filter (*Filter: ip.addr == 192.168.126.129*)

1. **What IP address was filtered?** The IP address 192.168.126.129 (my Kali Linux VM).
2. **Was the IP acting as source or destination?** It was acting as both. The ip.addr filter captures all traffic where the IP is either the sender (source) or the receiver (destination).
3. **What protocols were observed?** Various protocols such as TCP, TLS (from secure browsing), UDP, and DNS.

PART 2 — TCP Port Filter (*Filter: tcp.port == 443*)

1. **What service commonly uses port 443?** Port 443 is commonly used for HTTPS.
2. **Was the traffic encrypted?** Yes, the application data payload is encrypted using TLS.
3. **What websites generated the traffic?** Websites like google.com, youtube.com, and github.com generated this traffic.

PART 3 — Protocol Filter (*Filter: http*)

1. **Was the traffic readable?** Yes, when following the TCP stream, the HTTP traffic (requests in red, responses in blue) was completely readable in plaintext.
2. **Why is HTTP insecure?** HTTP is insecure because the data is not encrypted. Anyone who captures the packets can read the exact contents.
3. **What information was exposed?** Exposed information includes visited URLs, HTTP methods (GET/POST), Host headers, User-Agent details, and potentially sensitive data like login credentials if sent via POST.

PART 4 — Source and Destination Filter (*Filter: ip.src == 192.168.126.129 && ip.dst == 142.250.190.14*)

1. **Which device initiated communication?** My Kali VM (192.168.126.129) initiated the communication as the source device.
2. **What protocol was used?** TCP and TLS were primarily used for this communication.
3. **Were responses included?** No. The filter strictly specifies packets where Google is the destination.

PART 5 — MAC Address Filter (*Filter: eth.addr == 00:0c:29:45:c5:b2*)

1. **What is a MAC address?** A MAC (Media Access Control) address is a unique hardware identifier permanently assigned to a network interface controller (NIC) by the manufacturer.

2. **Is it different from an IP address?** Yes. A MAC address operates at Layer 2 (Data Link) and is used for local hardware delivery, whereas an IP address operates at Layer 3 (Network) and is used for logical routing across different networks.
3. **What packets involved this MAC?** All traffic entering or leaving the machine's network interface, including ARP broadcasts, IPv4, and IPv6 packets.

PART 6 — Conversations Filter (*Filter: (ip.addr == 192.168.126.129 && ip.addr == 142.250.190.14) && tcp.port == 443*)

1. **What service uses port 443?** HTTPS.
2. **Was the payload encrypted?** Yes, the payload was encrypted under TLS.
3. **What protocols appeared?** TCP (for the connection handshake and acknowledgments) and TLS (for the encrypted application data).

PART 7 — Exclude Filter (*Filter: !arp*)

1. **What protocol was excluded?** The ARP (Address Resolution Protocol) was excluded.
2. **Why remove ARP traffic?** ARP operates via constant network broadcasts, which can be very noisy. Removing it declutters the capture interface.
3. **Did analysis become easier?** Yes, filtering out the noise makes it much easier to focus on actual application traffic like TCP, DNS, and HTTP/S.

PART 8 — TCP Flags Filter (*Filter: tcp.flags.syn == 1 && tcp.flags.ack == 0*)

1. **What is a SYN packet?** A SYN packet is the first step in the TCP 3-way handshake, used to initiate a connection.
2. **Why is SYN important?** It is required to establish a reliable connection between a client and a server before any actual data can be transmitted.
3. **How can attackers abuse SYN packets?** Attackers can launch SYN Flood attacks sending thousands of SYN requests without completing the handshake to exhaust server resources.

PART 9 — DNS Queries Filter (*Filter: dns.flags.response == 0*)

1. **What is DNS?** Translates human-readable domain names into numerical IP addresses.
2. **Why are DNS queries important?** Acting a clear indicator of user intent.
3. **What domains were requested?** Domains like google.com, youtube.com, and cnn.com.

PART 10 — Packet Length Filter (*Filter: frame.len > 1000*)

1. **Why are some packets large?** They are carrying a maximum payload of application data.
2. **What applications generate large packets?** Video streaming (YouTube, Netflix).
3. **Could large packets indicate file transfers?** Yes. A continuous stream of large packets is a strong bulk data transfer.